

Speech Recognition

Exercises 05 NLTK 1

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Prerequisites

For these exercises it is recommended to:

- Work through ch01 and ch02 of the book <https://www.nltk.org/book>
- A good tutorial can also be found at Kaggle: <https://www.kaggle.com/code/alvations/basic-nlp-with-nltk>

In Python you will need:

- The NLTK ist installed.
- Download and install the corpora gutenber, web, brown (see chapter 02 1.1, 1.2, 1.3)
- some imports like:
 - `from nltk import word_tokenize`
 - `from urllib import request`

Part 01

Use the corpus module and the gutenber corpus to explore the text `austen - persuasion . txt`.

Implement the function from the slide 12 (ch01 in the NLTK book).

What is the lexical richness / diversity of the text `austen - persuasion . txt`?

Part 02

What does the following Python code do?

```
sum(len(w) for w in text1)
```

Use it to calculate the average word length of a text. Write a function for this task.

Part 03

Write a function `word_freq()` that takes a word and the name of a section/category of the Brown Corpus as arguments. It must compute the frequency of the word in that section of the corpus.

Part 04

Rewrite the function from part 03 so it can work with `text5` ("chat corpus").

Use the new function from part 03 and take an often used word of `text5` that is not a stop word.

Find the concordances, similarities, collocations in text5 of the NLTK book text examples (after from `nltk .book import *`).

Part 05

Write expressions for finding all words in text6 that meet the conditions listed below. The result should be in the form of a list of words: `['word1', 'word2', ...]`.

- Ending in `ise`
- Containing the letter `z`
- Containing the sequence of letters `pt`
- Having all lowercase letters except for an initial capital (i.e., `titlecase`)

Part 06

Analyze the language of a webpage and compare it with the results from exercises part 01 and 02.

Calculate the average word length and lexical richness.

As an example, you may take a THD Blog Artikel like

<https://www.th-deg.de/de/Blog/Artikel?id=2216430>